Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

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***Abstract*—Melanoma classifiers can exploit acquisition artifacts, and a prototype network's explanation may then point at them. On ISIC 2019, dark corners and bright structures remain associated with melanoma after stratifying by source. Mask supervision steers where prototype networks look, but, to our knowledge, no dermoscopic method controls which training patches the prototypes become. We propose lesion-anchored prototype learning: a lesion-attraction objective on each prototype's normalized spatial attention, and a projection that admits only cells at least half lesion and nearly free of detected artifacts. Every prototype thus comes from a lesion-dominated patch by construction, although its receptive field extends beyond it. In a same-backbone, leakage-audited comparison, the strongest mask-supervised ProtoPNet reaches similar accuracy and, across seven runs, robustness and generalizes better to Derm7pt, but puts all prototypes on lesion-dominated cells in only one of seven runs and learns more redundant prototypes. Against ProtoPNet, lesion attraction raises the AUC from 0.825 to 0.865 (three-seed means; 0.866 to 0.884 for seed ensembles), also with the background masked, makes the model less sensitive to swapped backgrounds and roughly halves decision flips under dark corners inserted around bit-identical lesion pixels (10.9%→5.2%), across seven runs per model. Code and logs: https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], yet they also learn *shortcuts* [2]: features correlated with the label in the training data but unrelated to the disease. In dermoscopy, surgical skin markings raise the melanoma probability of a deep classifier [3], models perform far above chance even when the lesion is removed [4], [5], and dark corners, rulers, gel bubbles and ink are common and unevenly spread across sites [6]–[8].

Rudin [9] argues that clinical use calls for models interpretable by design. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so a decision reads "this part looks like that part". Yet a prototype may be a piece of a ruler or a black corner; the shortcut then survives, presented as a reasonable explanation [11], as seen in dermoscopy [12], [13].

Research gap. Three gaps remain (Table I). First, dermoscopic debiasing methods [7], [14], [15] improve robustness to artifacts but are black boxes. Second, in prototype networks, mask supervision penalizes activation outside annotated regions [12], [16], and user feedback removes or deactivates unwanted prototypes [13], [17]. None constrains the *push* step, which decides which training patch a prototype finally *is*; feedback helps only after a user spots the problem. Third, artifact insertion has tested the *decisions* of classifiers [13], [15], [18], but, to our knowledge, not whether a prototype network's *explanations* stay on the lesion when artifacts are added or removed. **Contributions.** We address these gaps:

- **Lesion attraction on the normalized prototype attention.** An objective on each prototype's spatial attention pulls it toward the lesion without shrinking activations and accounts for the gains: without it, accuracy drops and flip rates under dark corners, rulers and ink roughly double; with it, ProtoPNet gains 4.0 AUC points (three-seed means; 1.8 for seed ensembles; 3.4 with the two core components alone). Across runs, the main model is on par with the closest prior work [12] in accuracy and robustness (Section VI-G), with a more stable artifact share and less redundant prototypes.
- **Lesion- and artifact-constrained projection.** The push step admits only latent cells whose input patch is at least half lesion and at most 5% detected artifact, so every prototype comes from one *by construction*. Mask losses give no such guarantee (all 20 prototypes on such cells in only one of 19 mask-loss runs). In 12 networks projected both ways, it fixes which patch an explanation shows without consistently changing AUC, flips or maps. Unlike a concurrent foreground-restricted projection [19], it also excludes artifacts and is tested against shortcuts.
- **A counterfactual, leakage-controlled test of decisions *and* explanations.** We insert four artifact types and remove real hair and ruler marks with every lesion pixel bit-identical and measure how far decision and explanation move, with a null control, unedited images, background masking and swaps and a detector-free audit. We group near-duplicates with each lesion, validate every mask source and the artifact detectors, and compare models sharing backbone and split with lesion-grouped statistics,

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TABLE I

POSITIONING AGAINST RELATED WORK. ANNOT.: HUMAN INPUT; INTERP.: INTERPRETABLE BY DESIGN; WHERE: CONSTRAINS WHERE THE MODEL LOOKS; PROV.: CONSTRAINS WHICH TRAINING PATCHES PROTOTYPES BECOME; CF: ARTIFACT INTERVENTIONS TESTING DECISIONS (D) AND EXPLANATIONS (E). †APPENDIX VARIANT.

Method	Annot.	Interp.	Where	Prov.	CF
EPVT [7], Bevan [14]	image labels	–	–	–	–
R2R [15]	semi-auto.	–	✓	–	D, E
RRR [21], CDEP [22]	pixel masks	–	✓	–	–
ProtoPNet [10]	none	✓	–	–	–
IAIA-BL [16]	expert masks	✓	✓	–	–
Correia *et al.* [12]	auto. masks	✓	✓	–	–
ProtoPDebug [17]	user	✓	✓	post hoc	–
PIP-Net (med.) [13]	user	✓	–	post hoc	D
Lin, Wright [19]	object masks	✓	✓	foreground†	–
Ours	auto. masks	✓	✓	**les., no art.**	**D, E**

across runs and on three external sets.

The main model, fixed before testing, keeps an auxiliary artifact-suppression term; alone it brought no significant gain, so we report it as a largely negative result, not a contribution, and report Ours-noArt alongside.

II. RELATED WORK

Artifacts and shortcuts in dermoscopy. Bissoto *et al.* annotated seven artifact types, found that simple debiasing does not remove their bias [5] and built artifact-based generalization benchmarks [6]. Removing dark corners barely changes accuracy but moves activations onto the lesion [8], and pasted colored patches make a CNN call malignant lesions benign [18]. EPVT [7] learns domain prompts for artifact groups, Bevan and Atapour-Abarghouei [14] unlearn artifact information adversarially, and Reveal-to-Revise (R2R) [15] finds band-aids, rulers and skin markers semi-automatically, corrects the model with RRR, CDEP or ClArC and tests it with pasted artifacts. All are black boxes, so we compare within prototype networks; they could complement anchoring.

Prototype networks and explanation supervision. The similarity a prototype network suggests can be misleading [11], so explanations should be tested instead of taken at face value [20]. RRR [21] penalizes input gradients inside annotated irrelevant regions, and CDEP [22] penalizes contextual-decomposition importance, demonstrated on ISIC images with colored patches. IAIA-BL [16] adds a fine-annotation loss on prototype activation outside regions marked by radiologists. Closest to us, Correia *et al.* [12] apply this loss to melanoma-versus-nevus classification with lesion masks from a segmentation network; our PPN-FA baselines follow it, in its original and a weaker formulation (Section VI-G). ProtoPDebug [17] and a medical PIP-Net study [13] let users forget or switch off confounded prototypes. Concurrently, Lin and Wright [19] study architectural prototype diversity on birds and faces, with a hinge loss on the background share of each prototype's activation and, in an appendix, a projection restricted to patches with more than 50% foreground.

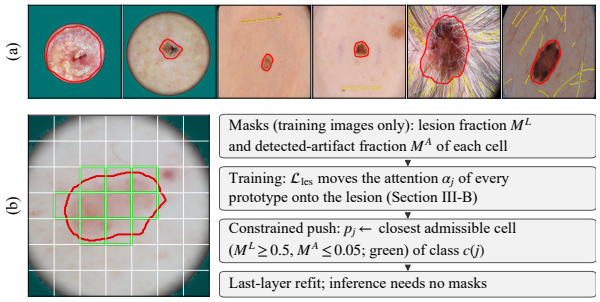


Fig. 1. (a) Automatic masks on ISIC 2019 images (red: lesion; cyan: dark corners; yellow: rulers, third and fourth, and hair, last two); artifact masks never overlap the lesion. (b) Method overview: the 7×7 cells of a test image that the constrained projection would admit (green), and the pipeline.

III. METHOD

A. Prototypical part network

Our network uses a DenseNet-121 backbone [23] (Fig. 1(b)). Two 1×1 convolutions and a sigmoid give a latent map $z \in [0,1]^{H \times W \times D}$, $H = W = 7$, $D = 128$. The $m = 20$ prototypes $p_j \in \mathbb{R}^D$ are ten per class, $c(j)$ being the class of p_j. Each cell has distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_c \mathcal{L}_{\text{clst}} - \lambda_s \mathcal{L}_{\text{sep}} + \lambda_l \mathcal{L}_{\text{les}} + \lambda_a \mathcal{L}_{\text{art}}, \quad (1)$$

where $\mathcal{L}_{\text{CE}}$ is a class-weighted cross-entropy and the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10] ($\lambda_c = 0.8$, $\lambda_s = 0.08$ as originally). Our model combines its two core components, lesion attraction $\mathcal{L}_{\text{les}}$ ($\lambda_l = 1$) and the constrained projection (Section III-C), with an auxiliary artifact-suppression term $\mathcal{L}_{\text{art}}$ ($\lambda_a = 1$; Section VI-H), all using the masks of Section IV.

B. Lesion attraction (core component)

Masks are augmented geometrically like the image, and average pooling to the 7×7 grid gives the lesion and artifact fractions of each cell, M^L and M^A. Where a prototype looks is given by its spatial attention $\alpha_j = \text{softmax}_{h,w}(s_j/\tau)$ with $\tau = 0.5$, a soft version of the max-pooling location behind a_j. Each prototype is weighted by $\omega_j = a_j / \sum_k a_k$, treated as a constant (stop-gradient), so that the prototypes the model uses count most. With $L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$, the attention mass of prototype j on the lesion of image i,

$$\mathcal{L}_{\text{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \quad (2)$$

where $\tilde{\omega}$ renormalizes ω over the prototypes of the true class y_i, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the batch images whose crop still shows lesion. The logarithm penalizes prototypes that almost ignore the lesion far more than those already on it. Since α_j sums to one, a larger lesion mass leaves less for everything else, artifacts included. The IAIA-BL fine-annotation loss [12], [16] penalizes activation magnitude outside a mask, whereas $\mathcal{L}_{\text{les}}$ acts on the normalized attention and leaves the activation

scale to the other losses. Unlike the concurrent hinge on the background activation share [19], it rewards the lesion mass logarithmically and weights each prototype by its use.

C. Lesion- and artifact-constrained projection (core component)

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of its class. This patch may lie on a ruler, a black corner or healthy skin even if training placed the attention well. We therefore admit only cells with $M^L \geq 0.5$ and $M^A \leq 0.05$ (Fig. 1(b)); no class ever lacked admissible cells. By construction, the "that part" of each explanation is then a 32×32 input patch that is at least half lesion and at most 5% detected artifact. The constraint fixes where a prototype comes from, not every pixel that shapes its latent vector, since a cell also sees its surroundings (Section VI-C). After the push, only $\mathbf{W}$ is trained (backbone, add-on layers and batch-normalization statistics fixed), so each prototype stays identical to its source patch. Five last-layer epochs did not solve this convex problem for *any* prototype model, so we re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an ℓ_1 penalty of 10^{-4} on negative-class connections [10]; 3,000 full-batch Adam steps, cosine-annealed learning rate from 0.02), identically for every prototype model and without validation data; Section VI-B also reports results without it.

IV. Masks: Provenance, Reliability and Leakage Control

A. Provenance and validation of the lesion masks

Each image is tagged with its mask source. For the 10,015 HAM10000 images (39.5% of ISIC 2019) we use the public expert segmentations [24] unchanged. The 15,316 BCN20000 [25] and MSK [26] images have no public masks; we predict them with a DeepLabV3 [27] network with an ImageNet-initialized MobileNetV3-Large [28] encoder trained at 256×256 (12 epochs, AdamW, one-cycle learning rate up to 5×10^{-4}, binary cross-entropy plus soft Dice, random flips and 90° rotations) *only* on the 8,163 HAM10000 images of our (previously fixed) training split. Predictions are averaged with those of the flipped image, thresholded at 0.5, reduced to the largest component and hole-filled. Masks covering less than 0.5% of the image (19 images, 0.08%) are marked invalid and excluded from mask-based losses and metrics. On the 1,852 HAM10000 images of the validation and test splits, which the segmentation network never saw, predicted masks reach a mean Dice of 0.9445 against the expert masks (median 0.9672, mean Jaccard 0.9015), and on the 200 expert-segmented PH2 images [29], from another clinic and device, 0.918 without adaptation. Section VI-I tests the dependence on mask quality.

B. Artifact masks

Artifact masks come from fixed, label-free image operations at 256×256 and always exclude the dilated lesion. *Dark corners* (DC) are dark regions (HSV value < 50) touching the border and at least 8 px thick on average. *Thin dark structures* (TS: hair, ruler ticks) are found with a 13×13 black-hat filter as in DullRazor [30] plus an elongation test, *bright structures* (SP: specular highlights, gel-bubble rims, also light hairs) with an 11×11 white top-hat filter, and *ink or colored patches* (INK) as strongly saturated blue-to-violet pixels (Fig. 1(a)). An image is flagged once a mask covers more than 1%, 0.3%, 0.05% or 0.1% of it. Against the public manual labels of Bissoto *et al.* [5] for 2,593 of our MSK images, the flags are precise but conservative (precision/recall: DC vs. dark corner 0.92/0.10, TS vs. hair or ruler 0.96/0.62, SP vs. gel bubble or border 0.86/0.35, INK vs. ink or patches 0.74/0.25); a visual check of 24 random DC and 24 random TS flags agrees (24 and 20 correct). Many artifacts are missed, above all faint dark corners (37% vs. 4% of these images). "Without artifacts" therefore means "without *detected* artifacts": an undetected artifact can enter a prototype cell, artifact shares count detected ones only, and the reference group of Section VI-E contains undetected ones, pulling A toward 0.5 for every model. In the 2,092 manually labeled training images, only 1.9% of the admissible cells lie on the outer ring of an image with an undetected dark corner. For hair or rulers, gel and ink, which the labels do not localize, only loose upper bounds exist (40%, 43%, 8%). None of the 5 prototypes of Ours from these images is such a cell.

C. How masks are used, and why they cannot leak

Masks serve only the training losses, the constrained projection, the attribution metrics and the placement of edits; the classifier never receives one, so test predictions do not depend on test masks. The split was computed first, from identifiers and image content alone, before any model was trained. Grouping by lesion identifier and near-duplicate pairs (Section V-A) can only remove leakage; an audit found no group, lesion identifier or image shared between sets, and all 342 near-identical pairs (thumbnail correlation ≥ 0.97) and other verified near-duplicate pairs lie within one set. Supervision uses training data only (segmentation network, mask-based losses, projection, last-layer re-optimization). Thresholds and checkpoints are chosen on validation, and the 1,852 expert masks of validation and test serve only for evaluation. Neither the segmentation network nor the artifact detectors see a diagnosis, so the masks cannot carry label information. Derm7pt, PAD-UFES-20 and PH2 are used for evaluation only; removing external images flagged as possible near-duplicates of ISIC training images (8, 6, 7) changes no external AUC by over 0.004.

V. Experimental Setup

A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [24], BCN20000 [25], MSK [26]; 4,522 melanomas, 17.9%), resized to 256×256. Unlike common practice, we keep the 2,084 MSK images without a lesion identifier. To avoid leakage, we group images by lesion identifier and by near-duplicate pairs (perceptual-hash distance ≤ 8 and thumbnail correlation ≥ 0.9 for full and central image). The 13,259 groups are split into ten stratified folds, one for testing and one for validation

TABLE II

THE ISIC 2019 SPLIT AND THE PREVALENCE OF DETECTED ARTIFACTS. LESION MASKS: EXPERT FOR HAM10000, PREDICTED FOR BCN20000 AND MSK (VALID FOR 99.91, 99.96 AND 100% OF THE TRAINING, VALIDATION AND TEST IMAGES). EXTERNAL SETS: DERM7PT, 1,011 CASES WITH A DERMOSCOPIC AND A CLINICAL IMAGE (252 MELANOMAS); PAD-UFES-20, 2,298 IMAGES (52 MELANOMAS); PH2, 200 IMAGES (SEGMENTATION ONLY).

	Train	Val.	Test	All
Images (% mel.)	20,552 (17.8)	2,459 (19.3)	2,320 (16.8)	25,331 (17.9)
Diagnosis				
Melanoma	3,657	475	390	4,522
Nevus	10,523	1,208	1,144	12,875
Basal cell carc.	2,671	322	330	3,323
Actinic keratosis	693	80	94	867
Benign keratosis	2,095	278	251	2,624
Dermatofibroma	193	16	30	239
Vascular lesion	201	27	25	253
Squamous cell carc.	519	53	56	628
Source (images / melanomas)				
BCN20000	10,037/2,299	1,225/310	1,151/248	12,413/2,857
HAM10000	8,163/917	952/103	900/93	10,015/1,113
MSK	2,352/441	282/62	269/49	2,903/552
Artifact prevalence (%)				
Dark corners	25.5	25.7	25.1	–
Thin dark structures	24.2	25.7	23.4	–
Bright structures	13.8	14.4	13.3	–
Ink / patches	1.6	1.9	2.0	–

(about 81/10/9% per source), with almost identical composition (Table II). External test sets are Derm7pt [31] (melanomas include *in situ* lesions and metastases) and the smartphone images of PAD-UFES-20 [32]; PH2 lacks diagnoses in our copy and only checks the segmentation (Table II).

B. Models and training

All configurations share backbone (ImageNet-pretrained DenseNet-121), input size (224×224), data, split and schedule. **CNN**: backbone, global average pooling, linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [16] on our lesion masks as in Correia *et al.* [12], with their $\lambda = 10^{-3}$ but computed on the 7×7 maps and averaged over the batch, which weakens it (Section VI-G tests the original formulations and larger weights). **Ours**: lesion attraction, constrained projection and the auxiliary $\mathcal{L}_{\mathrm{art}}$ ($\lambda_l = \lambda_a = 1$ in Eq. (1)). **Ablations**: without $\mathcal{L}_{\mathrm{les}}$ (Ours-noLes), without $\mathcal{L}_{\mathrm{art}}$ (Ours-noArt, the two core components only), with the standard projection (Ours-stdPush).

We train with Adam, batch size 64 and mixed precision: 5 epochs with frozen backbone (add-on layers and prototypes at 3×10^{-3}), 15 epochs of all layers (backbone 10^{-4}, weight decay 10^{-3}, learning rate divided by 10 every 5 epochs), then the projection and 5 last-layer epochs. The CNN trains for the same 20 epochs and keeps its best validation checkpoint. Class weights are 0.608 and 2.81. Prototype models are always evaluated after the projection, with the last-layer epoch chosen on validation. Each configuration runs with seeds 0, 1 and 2 (about 30 min per run on an NVIDIA T4); PPN, Correia (Section VI-G) and Ours also have seeds 3–6, with trained prototypes stored before the projection. Images and masks, stored at 256×256, receive the same random crop and flips before bilinear resampling to the 224×224 input; test images are only resized. After pooling to the 7×7 grid, each cell covers a 32×32 input patch, and attribution maps are compared with masks at 224×224. Loss weights and the eight configurations, including the choice of the main model, were fixed before any test evaluation; later variants and seeds changed none of them.

C. Evaluation protocol

Classification. Each run's threshold is set on ISIC validation (Youden index) and reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are mean $\pm$ std over seeds; tests use seed ensembles (mean probability).

Attribution. For a prototype model, we upsample the similarity maps of the three most activated prototypes to 224×224; their top 5% of pixels (the 95th-percentile region that ProtoPNet visualizes) form the hot region. For the CNN we use the class-activation map, which with a global-average-pooling head equals Grad-CAM [33] up to a positive factor [34]. We report the share of the hot region on the lesion and on artifacts, whether its maximum falls on an artifact (pointing), and the *enrichment*, i.e., artifact share divided by artifact area (above 1: more attention than the area explains).

Counterfactuals. Outside a 4-pixel margin around the lesion, we add a ruler along one border, 12–29 curved hairs, a dark vignette or 2–4 purple ink arcs to each test image. Conversely, detected thin dark structures (real hair and ruler marks, 544 images) are erased by Telea inpainting [35] outside a 2-pixel margin. As a null control, a same-shaped region on artifact-free skin is inpainted. We verify that every lesion pixel stays bit-identical and record the mean $|\Delta p|$ of the melanoma probability, the rate of flipped decisions and, for the three prototypes most active on the original image, the Spearman correlation ρ between the top-1 maps before and after the edit and the *capture*, the new hot region's share on inserted pixels. Lesion-only images keep the lesion and a 7-px perilesional rim and set all other pixels to the median color of a 10-px skin ring; background-only images Telea-inpaint the dilated lesion. Background swaps [36] paste each test lesion, pixel-identical and blended across its rim, onto the background-only image of a test image from another lesion group, drawn once (fixed seed, sources unmatched) from the same class (Mixed-Same), any class (Mixed-Rand) or the opposite class (Mixed-Opp), with identical compositing; BG-Gap $= \mathrm{AUC}_{\mathrm{Same}} - \mathrm{AUC}_{\mathrm{Rand}}$.

Statistics. The 2,320 test images form 1,234 groups (a lesion with its near-duplicates; 471 with several images), so we resample groups: ensemble AUCs get 95% CIs from 1,000 group resamples and are compared by a paired bootstrap with 2,000 (two-sided p), per-image metrics (averaged over seeds, then within groups) by the paired Wilcoxon test. Derm7pt (one image per case) uses the paired DeLong test [37], [38], PAD-UFES-20 a bootstrap over lesions. The Holm-corrected primary family holds the 16 main-claim tests: Ours against each of PPN-std, PPN, PPN-FA and the strongest mask-loss

TABLE III
ARTIFACT–MELANOMA ASSOCIATION (TRAINING SPLIT). OR: ODDS RATIO; MH: MANTEL–HAENSZEL OR STRATIFIED BY SOURCE.

Artifact	Prev.	$P(\mathrm{m}\mid\mathrm{a})$	$P(\mathrm{m}\mid\neg\mathrm{a})$	OR	MH OR [95% CI]
Dark corners	25.5%	.263	.149	2.03	1.555 [1.43, 1.69]
Bright structures	13.8%	.266	.164	1.85	1.751 [1.59, 1.92]
Thin dark struct.	24.2%	.174	.179	0.97	0.988 [0.90, 1.08]
Ink / patches	1.6%	.202	.178	1.18	1.114 [0.85, 1.46]
Any	48.8%	.210	.148	1.53	1.265 [1.17, 1.36]

variant of Section VI-G on AUC, artifact share and flip rate under dark corners and rulers. All other tests are secondary and uncorrected; of the 12 main background, diversity and coverage comparisons, 10 survive Holm correction (not lesion-only ensembles vs. PPN-FA, IAIA-BL: $p = 0.06$). Image-level tests agree unless noted. A run is one trained network; tables average seeds 0–2, ensembles average their probabilities, ensemble and per-image tests treat models as fixed, and across-run tests (exact Mann–Whitney) use all runs (seven for PPN, Correia and Ours). Pre-specified outcomes were BG-Gap, coverage of the non-melanoma diagnoses, background-only AUC, prototypes with $d > 0.5$ under dark corners, criterion-aligned prototypes (Section VI-D) and the excess of the on-lesion hot region on expert-annotated dermoscopic structures over their lesion share [39] (501 validation and test images with ISIC 2018 Task 2 masks). p-values below 10^{-6} are given as $p < 10^{-6}$.

VI. RESULTS

Main models: CNN, PPN-std, PPN, PPN-FA, Ours (others: Sections VI-F–VI-H); bold: best main prototype model.

A. Artifacts are associated with melanoma

In training images, melanomas show dark corners and bright structures significantly more often than other lesions (Table III). Source could explain this (dark corners: BCN 47%, HAM 5%; BCN has the highest melanoma rate), but source-stratified Mantel–Haenszel ORs stay above 1: for dark corners in every source (BCN 1.41, HAM 3.10, MSK 2.24), for bright structures in BCN and HAM (2.11, 1.92) but not MSK (0.88). Strength differs by source (Breslow–Day $p < 10^{-6}$), so pooled ORs are averages; thin dark structures and ink are not associated.

B. Classification

On ISIC test, Ours has the highest AUC of the main prototype models (Table IV). Its ensemble exceeds those of PPN-std, PPN and PPN-FA by 0.022, 0.018 and 0.017 (95% CI $[0.009, 0.036]$, $[0.004, 0.032]$, $[0.005, 0.031]$; Holm-adjusted $p = 0.012$, 0.042, 0.029), and its seeds vary less. Each of the three Ours seeds beats each of the 13 seeds of PPN-std, PPN (seven) and PPN-FA (worst 0.8592 vs. best 0.8526; Derm7pt-dermoscopic 0.7405 vs. 0.7343). One of seven Ours seeds (0.863±0.012, range 0.842–0.880) falls below the best baseline seed; across runs, Ours exceeds these 13 seeds and the seven PPN seeds (0.821±0.039; exact Mann–Whitney $p = 0.001$, 0.007). At the validation threshold, Ours trades sensitivity for specificity; at 90% sensitivity (test ROC), its specificity is highest among prototype models (64.8%; PPN-std, PPN, PPN-FA 58.9–60.2%, Correia 63.3%, CNN 69.6%; ensembles $p \geq 0.14$; over seven runs vs. PPN $p = 0.025$, vs. Correia $p = 0.90$), and a validation threshold for 90% sensitivity reaches 92.6% on test. The black-box CNN remains more accurate ($p = 0.009$), a price of about 3 ensemble AUC points for interpretability (4.6 for PPN) that vanishes with random backgrounds (below). With only the five last-layer epochs, all prototype models score lower, Ours least (0.019 vs. 0.039–0.079), and it stays best (0.846±0.007; baselines 0.765–0.797, ablations 0.787–0.841; $p \leq 0.001$ against the baselines).

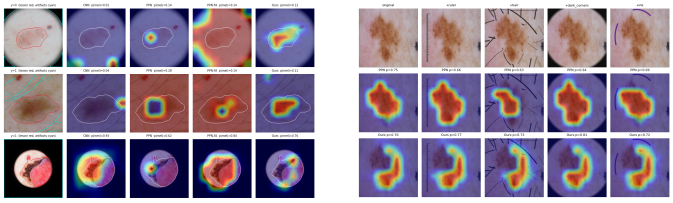


Fig. 2. (a) ISIC test images with artifacts (lesion red, detected artifacts cyan); CNN: class-activation map; others: map of the most activated prototype; p: melanoma probability (seed 0). (b) Counterfactual (lesion pixels unchanged; second of the first two test images receiving a ruler, not selected); rows 2–3: map of the prototype most active on the original, and p, for PPN and Ours; only the map of PPN shifts with added hair.

Lesion versus background. On lesion-only images, the Ours ensemble keeps its lead over PPN-std, PPN, PPN-FA and IAIA-BL (0.874 vs. 0.850–0.859, $p \leq 0.03$; Correia 0.878, n.s.), and its seven runs exceed all 13 ProtoPNet runs and those of Ours-noLes ($p = 3 \times 10^{-5}$, 0.017); Ours and the prototype baselines lose at most 0.011 AUC, the CNN 0.024. On background-only images, the ensembles still reach 0.61–0.67 (Ours 0.635, $p \geq 0.15$; across runs PPN scatters around chance, 0.44–0.66, $p = 0.011$), but lesion size alone, still visible as the filled region, matches this (0.67), so only the swaps, which keep the lesion fixed, isolate the background. Compositing alone costs every model at most 0.022 AUC (Mixed-Same vs. unedited, mean over runs; Ours 0.857 vs. 0.863). The BG-Gap of the Ours ensemble is about half that of PPN-std, PPN, PPN-FA, the CNN and Ours-noLes (0.023 vs. 0.039–0.059; upper 95% CI bounds of the differences ≤ -0.005, $p \leq 0.002$; across runs $p = 0.015$ vs. 13 ProtoPNet runs), on par with the mask losses (0.018, $p \geq 0.22$). On random backgrounds Ours is as accurate as the CNN (mean over runs 0.834 vs. 0.826), on opposite-class ones nominally more (0.803 vs. 0.778, $p = 0.067$), so part of the CNN's lead may rest on the background.

Externally, Ours has the highest AUC of the main prototype models on both Derm7pt sets. On the dermoscopic images its ensemble beats PPN-FA and PPN-std ($p = 1.7 \times 10^{-5}$, 2.8×10^{-6}, also after Holm correction) but not PPN ($p = 0.20$), whose seeds vary strongly. On the clinical images, its advantage over PPN-std and PPN ($p = 0.023$, 0.046) does not survive Holm correction ($p = 0.069$, 0.092), and over PPN-FA is not significant ($p = 0.30$). On PAD-UFES-20 (52 melanoma images of 36 lesions), Ours beats PPN-std ($p = 0.001$) and does not differ significantly from PPN and PPN-FA ($p \geq 0.23$); its mean AUC there is even below that of PPN. The black-box EPVT reports 0.837 on Derm7pt-dermoscopic [7] (melanoma vs. benign, other split, ViT-B/16); our CNN reaches 0.799.

TABLE IV

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% CI; p: ENSEMBLE AGAINST OURS ON ISIC TEST (PAIRED BOOTSTRAP OVER LESION GROUPS). PPN-FA: IAIA-BL LOSS ON 7×7 MAPS (ORIGINAL FORMULATIONS IN TABLE VIII).

	ISIC 2019 test (n = 2,320)						External AUC		
Model	AUC	BAcc	Sens.	Spec.	Ens. AUC [95% CI]	p	Derm7pt-derm	Derm7pt-clin	PAD-UFES-20
CNN (black box)	.902±.006	.814±.011	.808±.044	.820±.040	.912 [.884, .935]	.009	.799±.023	.704±.022	.721±.022
PPN-std	.835±.013	.752±.014	.769±.063	.734±.040	.861 [.827, .888]	.002	.677±.027	.624±.023	.587±.019
PPN	.825±.040	.758±.010	**.786±.068**	.730±.087	.866 [.830, .894]	.014	.688±.072	.618±.038	**.665±.016**
PPN-FA (7×7)	.844±.008	.744±.021	.689±.045	.800±.005	.866 [.830, .895]	.007	.696±.024	.640±.013	.630±.059
Ours	**.865±.006**	**.780±.014**	.734±.075	**.825±.046**	**.884 [.850, .910]**	–	**.749±.009**	**.657±.019**	.647±.108
Ours-noLes	.830±.015	.746±.030	.766±.091	.726±.032	.856 [.821, .885]	<.001	.717±.016	.636±.033	.628±.047
Ours-noArt	.859±.011	.772±.003	.733±.056	.811±.054	.877 [.845, .904]	.22	.752±.014	.637±.030	.678±.018
Ours-stdPush	.863±.005	.785±.002	.823±.043	.747±.047	.884 [.852, .911]	.86	.747±.005	.670±.019	.658±.108

TABLE V

WHERE THE MODELS LOOK (MEAN OVER 3 SEEDS). LES., ART.: HOT-REGION SHARE ON THE LESION AND, IN IMAGES WITH ARTIFACTS, ON ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; PT.: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES ON CELLS WITH ≥50% LESION (OF 20; 20 BY CONSTRUCTION UNDER THE CONSTRAINED PROJECTION, NOT RANKED); DC–INK: SHARE ON THAT ARTIFACT IN IMAGES CONTAINING IT; D7: DERM7PT-DERMOSCOPIC.

	ISIC test					ISIC test, by artifact type				D7		
Model	Les.↑	Art.↓	Enr.↓	Pt.↓	L-pr.	DC	TS	SP	INK	Les.	Pt.	Enr.
CNN	.466	.231	1.52	.117	–	.365	.016	.017	.029	.605	.153	.63
PPN-std	.518	.082	0.54	.050	12.0	.123	**.010**	.015	.049	.749	.075	.25
PPN	.562	.066	0.46	.042	14.3	.098	**.010**	.012	.036	**.789**	.044	.24
PPN-FA	.530	.102	0.68	.050	10.0	.158	**.010**	**.010**	.056	.760	.077	.28
Ours	**.589**	**.048**	**0.37**	**.013**	20.0	**.067**	.011	.016	**.032**	.734	**.015**	**.18**
Ours-noLes	.565	.106	0.67	.053	20.0	.162	.011	.013	.044	.772	.057	.30
Ours-noArt	.594	.042	0.40	.018	20.0	.056	.012	.020	.040	.713	.017	.23
Ours-stdPush	.515	.086	0.58	.034	7.7	.129	.012	.019	.030	.633	.033	.35

C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table V, Fig. 2(a)): in images with artifacts, 23.1% of its hot region lies on them, 1.5 times their area share. The prototype models look at artifacts less than their area suggests, Ours least (4.8±1.5% of the hot region, maximum on an artifact in 1.3±0.5% of images; ±: std over seeds), with the largest lesion fraction. In every Ours run, all 20 prototypes lie on lesion cells ($M^L \geq 0.5$; Fig. 3), whereas PPN-FA projects on average 10.0 onto lesion and 2.0 onto artifact cells. Most attention on artifacts falls on dark corners, where Ours has the smallest share. On the test sample, Ours differs significantly from each other main model in lesion and artifact share, enrichment and pointing ($p \leq 5.6 \times 10^{-4}$; primary-family artifact shares: Holm-adjusted $p \leq 2.2 \times 10^{-5}$). Across runs, they overlap with PPN and PPN-std (Section VI-D). On Derm7pt-dermoscopic, it again has the lowest artifact share, enrichment and pointing ($p \leq 8.6 \times 10^{-3}$), but the prototype baselines place more hot region on the lesion ($p \leq 1.2 \times 10^{-4}$).

Receptive field of the prototypes. The effective receptive field (ERF) of a prototype's source cell is the input-gradient magnitude of its latent vector (eight random projections). For every model, only 8–12% of the mass lies inside the cell, and half spreads over 8–11% of the image. For Ours, 69±3% lies on the lesion (10th percentile 42%), vs. 48–53% for PPN-std, PPN and PPN-FA (10th percentiles 11–21%), and 2.5% on detected

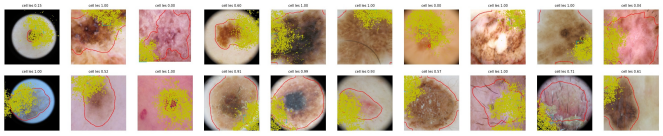


Fig. 3. Sources of the ten non-melanoma prototypes of PPN (top) and Ours (bottom), seed 0, with the projected cell (cyan; lesion fraction above each image), the lesion (red) and the smallest region holding half of the prototype's ERF (yellow). Four PPN prototypes come from background cells (lesion fraction 0.00–0.15) and draw on skin and vignette; all of Ours come from lesion cells (≥ 0.52, no detected artifact) and draw mostly, though not only, on the lesion.

artifacts (4.3%, 2.7%, 7.3%). This reflects where prototypes are taken, not a narrower field: the constraint selects source images whose lesion covers 54% of the image (baselines 40%), and relative to lesion area every model concentrates its ERF on the lesion by a median factor of only 1.2–1.3. Ours-stdPush drops to 47%; Ours-noLes, Ours-noArt keep 71–72%. For 13% of Ours prototypes (baselines 22–28%), the ERF draws less on the lesion than its area would, so anchoring covers the projected cell and most, not all, of its receptive field.

D. Counterfactual robustness

Table VI summarizes the interventions (2,063–2,318 images per insertion). Adding dark corners *increases* the melanoma probability of every model (mean signed change +0.041 CNN, +0.048 PPN, +0.020 Ours), as associated in Table III; Ours responds least. Rulers, hair and ink, not associated with melanoma (Table III), probe general robustness. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers, reduces it for ink and hair and flips less than the CNN under all four (Table VI). Against every other main model, all four reductions are significant ($p < 10^{-6}$ vs. the prototype baselines, also Holm-adjusted for dark corners and rulers; $p \leq 0.008$ vs. the CNN), as is the smaller $|\Delta p|$ vs. PPN ($p < 10^{-6}$) and, except for hair ($p = 0.09$), the CNN ($p \leq 2.3 \times 10^{-4}$). Removing real hair and ruler marks flips more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so inpainting cannot explain it.

Threshold and direction. At the validation threshold for 90% sensitivity, Ours still flips less than every baseline under dark corners (3.7% vs. 4.9–6.9%; $p \leq 0.004$), than the ProtoPNet baselines and the CNN under rulers (2.6% vs. 4.1–7.1%; $p \leq 0.002$) and than PPN-std, PPN, IAIA-BL and the

TABLE VI

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP (DECISION CHANGES), $|\Delta p|$, ρ AND CAP. (%) AS DEFINED IN SECTION V-C. THE NULL CONTROL IS NOT RANKED.

	+Dark corners		+Ruler		+Hair		+Ink		Remove hair/ruler		Null control		Expl.: dark corners		Expl.: ruler	
Model	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	$\rho\uparrow$	Cap.$\downarrow$	$\rho\uparrow$	Cap.$\downarrow$
CNN	6.1	.049	5.6	.049	12.5	.104	9.7	.080	4.7	.036	1.0	.008	–	–	–	–
PPN-std	10.9	.060	8.3	.041	14.9	.092	9.9	.058	8.8	.036	4.3	.013	.635	7.1	.773	1.5
PPN	10.9	.067	7.5	.045	13.9	.096	9.5	.062	7.4	.045	3.6	.019	.680	6.1	.804	1.1
PPN-FA	10.6	.049	6.5	.028	15.1	.084	9.1	.046	7.2	**.030**	3.2	.010	.714	11.4	.874	1.1
Ours	**5.2**	**.035**	**3.6**	**.025**	**9.2**	**.081**	**5.6**	**.045**	**4.4**	.032	1.6	.011	**.918**	**3.4**	**.952**	**0.7**
Ours-noLes	10.2	.053	8.7	.040	14.3	.083	10.1	.049	8.3	.042	3.9	.016	.705	8.6	.827	1.7
Ours-noArt	5.4	.036	3.5	.023	8.7	.061	6.1	.041	4.4	.027	1.0	.011	.884	3.4	.949	1.0
Ours-stdPush	4.5	.034	4.1	.028	9.6	.077	5.8	.045	4.5	.033	1.9	.011	.923	7.5	.944	1.4

TABLE VII

REAL, UNEDITED ARTIFACTS ON NON-MELANOMA TEST LESIONS. A: PROBABILITY THAT A NON-MELANOMA LESION WITH THE ARTIFACT SCORES HIGHER THAN ONE WITHOUT IT (SEED ENSEMBLES; 0.5: NO EFFECT); FPR: FALSE-POSITIVE RATE (%) WITH/WITHOUT THE ARTIFACT (VALIDATION THRESHOLD, MEAN OVER SEEDS). NON-MELANOMA IMAGES WITH/WITHOUT: 447/1,483 (ISIC, DC), 251/1,679 (ISIC, SP), 515/244 (DERM7PT-CLIN., DC).

	ISIC, dark corners		ISIC, bright struct.		Derm7pt-clin., DC	
Model	$A\uparrow$	FPR	$A\uparrow$	FPR	$A\uparrow$	FPR
CNN	.57	22/17	.58	19/18	.59	18/9
PPN-std	.58	31/25	.61	32/26	.82	60/21
PPN	.61	30/26	.60	33/26	.78	35/11
PPN-FA	.66	33/16	.62	26/19	.80	46/15
Ours	**.57**	**20/17**	**.59**	**19/17**	**.69**	**24/12**
Ours-noLes	.66	37/25	.63	37/26	.77	50/24
Ours-noArt	.58	20/19	.62	25/18	.75	37/12
Ours-stdPush	.57	28/25	.59	29/25	.65	33/19

CNN under ink ($p \leq 0.025$); under hair only than the CNN, and Correia flips fewer (9.8% vs. 11.4%, $p < 10^{-4}$). At the Youden threshold, dark corners and rulers mostly flip decisions to melanoma (PPN 9.7 of 10.9%, 5.2 of 7.5%), hair the reverse (Ours 7.7%, PPN 7.8%). Ours has fewer flips to melanoma than all prototype baselines under each insertion (corners, ruler, hair, ink: 4.1, 2.2, 1.5, 1.8% vs. $\geq$4.8, 2.5, 3.2, 2.6%).

The explanations of Ours also move the least (Fig. 2(b)). Under added dark corners, its top-1 map keeps $\rho = 0.92$ (PPN 0.68), and only 3.4% of its hot region (per seed 0.2–7.7%) moves onto them ($p < 10^{-6}$ vs. PPN and PPN-FA). Ink, drawn just outside the lesion border where anchored prototypes respond, is the exception: 1.8% of the hot region of Ours falls on it vs. 1.0% for PPN, though it still flips less.

Each of the seven Ours seeds flips fewer decisions under dark corners and ink than 12 of the 13 seeds of PPN-std, PPN and PPN-FA (3.3–7.5% vs. 7.6–15.4%; 4.4–8.0% vs. 8.0–11.2%, i.e., at most 184 vs. at least 185 of 2,301 images); the exception is the lowest-AUC PPN seed (0.752; 2.0%, 6.4%). Across runs, Ours flips less than these 13 seeds under all four insertions ($p \leq 0.004$) and the seven PPN seeds ($p \leq 0.026$), not the seven Correia seeds ($p \geq 0.62$). Every Ours seed also keeps a more stable map than any PPN-std or PPN-FA seed (ρ 0.85–0.93 vs. $\leq$0.77); ρ overlaps with two PPN seeds (0.87) and the mask-loss variants ($\leq$0.90). The artifact share (3.0–8.9%) lies below every CNN seed ($\geq$20.7%) but overlaps with the prototype baselines and Correia, and the captured share overlaps for all (Ours 0.2–7.7%, baselines 0.1–15.4%).

Prototypes without detectors. Under insertions, Ours' prototypes change their mean activation (d, in between-image SDs) as much as PPN's ($p \geq 0.62$), so their stability is not due to inertness. No baseline, IAIA-BL or Correia prototype reaches $d = 0.27$: flips stem from many small shifts set by training (Table X); single Ours prototypes rise more ($d \leq 0.84$; > 0.5 under hair in four runs, dark corners in one), yet it flips less. Ours has the least redundant prototypes: their activations on test images correlate less (mean absolute Spearman r 0.71 vs. 0.83–0.88), and the participation ratio $(\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ of the correlation matrix's eigenvalues λ_k (PR, effective number of distinct prototypes) is higher in every Ours run than in any baseline, IAIA-BL or Correia run (1.63–1.97 vs. 1.11–1.62; $p = 6 \times 10^{-4}$ vs. PPN, Correia), mainly among non-melanoma prototypes, via lesion attraction (Ours-noLes 1.41, Ours-stdPush 1.79), although one factor dominates in every model: the gain is relative, not semantic. The non-melanoma prototypes of Ours cover the non-melanoma diagnoses best: averaged over the seven, the best one-vs-rest AUC of such a prototype is 0.617 vs. 0.540–0.583 ($p = 0.002$ vs. PPN and Correia), although none reaches 0.75; the best of ten, chosen on test images, inflates all models alike. On Derm7pt, Correia aligns more prototypes with seven-point criteria (within-diagnosis AUC $\geq$0.60; 19 vs. 13, $p = 0.001$), but in fewer distinct groups ($r > 0.9$; 5.3 vs. 9.4, $p = 0.054$); all align weakly (mean best AUC 0.57–0.60). Of the pre-specified outcomes, BG-Gap and coverage favor Ours, the next three do not, and the excess on expert-annotated structures is null for every model (-0.009 to $+0.010$).

E. Shortcuts in unedited images

Without editing, we ask whether test-image artifacts raise the melanoma score of *non-melanoma* lesions (threshold-free A of Table VII). On ISIC, dark corners double the false-positive rate of PPN-FA but that of Ours only slightly. The effect A of Ours is significantly smaller than that of PPN, PPN-FA and Ours-noLes ($p < 0.001$) and does not differ from that of PPN-std and the CNN ($p \geq 0.77$). Lesion attraction thus reduces the real dark-corner effect below that of PPN and PPN-FA, though not below PPN-std or the CNN; our weak mask loss leaves it in place, and the original formulations also reduce it (Section VI-G). The same holds for bright structures, where Ours has the smallest effect of the prototype models ($p \leq$

TABLE VIII

MASK-LOSS VARIANTS (ISIC TEST UNLESS NOTED; MEAN OVER 3 SEEDS). CORREIA [12], IAIA-BL [16]: ORIGINAL FORMULATIONS (SECTION VI-G). D7-D: DERM7PT-DERMOSCOPIC AUC; L-PR., ART.: AS IN TABLE V; FLIP, ρ: INSERTED DARK CORNERS; A: REAL DARK CORNERS.

Mask loss	AUC	D7-d	L-pr.	Art.↓	Flip↓	$\rho\uparrow$	$A\downarrow$
None (PPN)	.825±.040	.688	14.3	.066	10.9	.68	.61
PPN-FA, 10^{-3}	.844±.008	.696	10.0	.102	10.6	.71	.66
PPN-FA, 10^{-2}	.833±.018	.695	16.0	.121	12.2	.80	.66
PPN-FA, 10^{-1}	.849±.006	.711	9.3	.061	7.2	.84	.60
Correia	.867±.011	**.771**	13.0	.141	6.5	.87	.57
IAIA-BL	.845±.011	.715	18.0	**.039**	6.2	.88	.58
Ours (no mask loss)	.865±.006	.749	20.0	.048	5.2	**.92**	.57
Ours+Correia	**.871±.004**	.736	20.0	.043	**5.0**	.87	**.53**

TABLE IX

ROBUSTNESS TO MASK QUALITY (ISIC TEST, MEAN OVER 3 SEEDS): ATTRIBUTION AS IN TABLE V ON IMAGES WITH EXPERT (HAM10000, $n = 900$) AND PREDICTED MASKS (BCN20000, MSK, $n = 1{,}420$); LESION SHARE WITH ALL LESION MASKS ERODED (-5) OR DILATED ($+5$) BY 5 PX; AUC PER SOURCE (TEST COUNTS IN TABLE II).

Model	Expert masks Les.↑	Art.↓	Pt.↓	Predicted masks Les.↑	Art.↓	Pt.↓	Lesion -5	$+5$	AUC by source BCN	HAM	MSK
CNN	.444	.043	.015	.480	.277	.182	.434	.496	.883	.934	.861
PPN-std	.416	.016	.004	.582	.098	.079	.480	.552	.825	.880	.711
PPN	.446	.014	.005	**.635**	.079	.065	.525	.595	.819	.860	.692
PPN-FA	.423	.019	.004	.599	.123	.080	.491	.567	.834	.896	.699
Ours	**.534**	**.013**	**.002**	.624	**.056**	**.021**	**.534**	**.640**	**.852**	**.906**	**.761**
Ours-noLes	.503	.023	.007	.603	.126	.082	.525	.601	.815	.879	.704
Ours-noArt	.543	.013	.003	.626	.050	.028	.541	.644	.853	.893	.737
Ours-stdPush	.459	.018	.003	.550	.103	.054	.466	.561	.852	.904	.749

0.038 against PPN-std and PPN-FA), and most clearly for dark borders in clinical Derm7pt images (DeLong $p \leq 1.4 \times 10^{-6}$). The CNN attends to dark corners most but has no larger effect.

F. Ablation of the two core components

Lesion attraction drives the accuracy gain. Without it (Ours-noLes), the test AUC falls to 0.830 ($p < 0.001$), the artifact share more than doubles (4.8%→10.6%), and dark-corner, ruler and ink flip rates roughly double. Ours-noLes keeps the constrained projection and all prototypes on lesion cells; a correct projection does not suffice without anchored attention.

The constrained projection determines what the prototypes *are* (L-pr. in Table V). It does not change the accuracy (ensemble difference within $[-0.006, 0.005]$, 95% CI) or, significantly, the flip rates ($p \geq 0.13$), and its map differences in Tables V and VI are not systematic. In a factorial control (Table X), we projected each network of seeds 3–6 of PPN, Correia and Ours both ways from its stored prototypes. In all 12 networks it put the 20 prototypes on lesion cells (before 6–17, one Correia network 20; mean lesion fraction 0.62→0.90; exact Wilcoxon $p = 0.001$) and changed no measure of accuracy, attribution or robustness consistently (AUC $+0.0001$; others $p \geq 0.067$, both signs); seeds 0 and 2 of Ours and Ours-stdPush, which share their trained network (seed 1 diverged), agree. With the same constrained projection, Ours keeps a higher ensemble AUC than PPN (0.886 vs. 0.866, $p = 0.001$), less artifact share ($p < 10^{-5}$) and fewer flips under all four insertions ($p \leq 4 \times 10^{-4}$), as with the standard projection: robustness comes from lesion attraction, not the projection.

TABLE X

FACTORIAL CONTROL: SEEDS 3–6 OF EACH TRAINING OBJECTIVE, EACH NETWORK PROJECTED BOTH WAYS (ISIC TEST; MEAN OF 4 NETWORKS). L-PR., ART.: AS IN TABLE V; FLIP: DECISION CHANGES (%) UNDER INSERTED ARTIFACTS; ρ: DARK CORNERS.

Loss	Push	L-pr.	AUC	Art.↓	Flip↓ DC	Ruler	Hair	Ink	$\rho\uparrow$
None (PPN)	std.	11.8	.818	.062	10.4	6.9	15.3	9.2	.75
	constr.	20.0	.819	.060	10.5	6.1	15.1	8.0	.81
Correia	std.	16.0	.860	.042	6.1	3.6	10.5	6.9	.90
	constr.	20.0	.860	.042	6.1	3.6	10.4	6.8	.90
Ours	std.	9.0	.861	.052	5.8	3.8	10.4	6.7	.89
	constr.	20.0	.861	.052	6.0	4.1	10.7	6.8	.89

On seeds 3–6 of Ours, lesion thresholds of 0.3–0.9 (artifact fraction $\leq$0.05) and artifact limits of 0, 0.2 or none (lesion fraction $\geq$0.5) put 14.8–20 of 20 prototypes on lesion cells (mean source-cell lesion fraction 0.78–0.99 vs. 0.47 unconstrained). The AUC (0.861–0.863), dark-corner flips (5.8–6.1%), artifact share (5.1–5.5%) and ρ (0.885–0.886) stay at the standard projection's (0.861, 5.8%, 5.2%, 0.886). Only the strictest (0.9) removes most border cells (2.3 vs. 8.5 at 0.5). The artifact limit itself excludes only 0.9% of training lesion cells (2 of the 20,333 training images with a lesion cell lose all).

Attraction strength. We retrained Ours with $\lambda_l = 0.1$, 0.3, 0.5, 2 and 3 (ten prototypes per class, three seeds each). From $\lambda_l = 0.1$ to 1, accuracy (AUC 0.848, 0.843, 0.857, 0.863), robustness (dark-corner flips 8.0, 9.1, 7.5, 5.8%) and prototype diversity rise (Spearman, 16 runs, $p \leq 0.030$ for AUC, PR and flips under dark corners, rulers and hair). At $\lambda_l \geq 2$, all runs reach AUC 0.89–0.90, but cluster and separation costs collapse ($\leq$0.02 and 0.23 vs. 1.8 and 50 at $\lambda_l = 1$; pre-set criterion: separation cost < 1; every run up to $\lambda_l = 1$: $\geq$47): patches match both classes' prototypes alike. Among non-collapsing values, validation AUC selects the pre-set $\lambda_l = 1$. In two seeds each, $\tau = 0.25$ matches Ours (AUC 0.864, flips 4.9%); $\tau = 1$ and 20 prototypes per class weaken the attraction per prototype and keep the AUC (0.856, 0.862) but flip more (9.6%, 9.5%), and five per class collapse (AUC 0.90, flips 5.9%).

G. Strength of the mask-loss baseline

IAIA-BL [16] and Correia *et al.* [12] compute the fine-annotation loss at 224×224 and sum it over the batch; with the same $\lambda = 10^{-3}$, it is about three orders of magnitude stronger than in PPN-FA (factors 32 and 64; Correia *et al.* penalize own-class prototypes only). We trained the original formulations and PPN-FA with $\lambda = 10^{-2}$ and 10^{-1}, three seeds each (Table VIII; secondary). Correia's formulation does not differ significantly from Ours in ISIC AUC (-0.001, 95% CI $[-0.012, 0.011]$) and generalizes better (Derm7pt-dermoscopic, -clinical, PAD-UFES-20: 0.771, 0.702, 0.699 vs. 0.749, 0.657, 0.647; $p \leq 0.028$), but puts fewer prototypes on lesion cells and three times more hot region on artifacts ($p < 10^{-6}$). In the primary family, Ours is significantly better on artifact share and dark-corner flips (Holm $p < 10^{-6}$, $p = 0.029$), not on AUC or ruler flips. Neither replicates on seeds 3–6: there, Correia puts 13–20 prototypes on lesion cells, has less artifact share (4.2% vs. 5.2%, $p < 0.001$) and flips as often ($p \geq 0.17$), while less

of the hot region of Ours moves onto inserted dark corners (1.0–1.9% vs. 3.8–4.4%, $p < 10^{-5}$); it again generalizes better to Derm7pt-clinical (0.716 vs. 0.689, $p \leq 0.022$). Over seven seeds, its artifact share spans 0.5–24.6% (Ours 3.0–8.9%), and neither AUC nor flip rates differ across runs ($p \geq 0.62$). IAIA-BL keeps its hot region on the lesion (83%) and off artifacts better ($p < 0.002$) but is less accurate (ensembles $p = 0.057$) with prototypes off lesion cells. On seeds 0–2, every variant flips more than Ours under dark corners and ink (6.6–10.7% vs. 5.6%; $p \leq 0.02$), though per seed the original formulations, and for dark corners $\lambda = 10^{-1}$, overlap with Ours; under rulers the original formulations ($p \geq 0.052$) and under hair Correia ($p = 0.37$) do not differ significantly, and none has as stable an explanation. The original formulations' real dark-corner effect does not differ significantly from that of Ours ($p \geq 0.26$), unlike PPN-FA. The two are complementary: mask supervision steers where the network looks and aids generalization.

Adding the Correia loss to Ours (Ours+Correia, three seeds) gives the highest ISIC AUC of all prototype models ($p \geq 0.32$ vs. Ours, Correia), maps more lesion-focused than Ours (lesion share 73% vs. 59%, seeds 59–85% vs. 49–69%; captured share 2.2% vs. 3.4%, less artifact share; $p < 10^{-6}$), the smallest real dark-corner effect of all models ($p < 0.001$ vs. both) and similar flip rates ($p \geq 0.26$; hair 7.9% vs. 9.2%, $p = 0.02$). It lacks the Derm7pt advantage of the mask loss (0.736 and 0.675 vs. 0.771 and 0.702; DeLong $p \leq 0.016$), although its maps are more lesion-focused than Correia's (73% vs. 60%), so lesion focus alone cannot explain it (untested: over-constrained attention, or seed variance); it has the highest PAD-UFES-20 AUC of all models (0.737, CNN 0.721; $p \leq 0.001$ against both) and less stable maps than Ours.

H. Artifact suppression: a largely negative result

The auxiliary term penalizes artifact attention directly: with $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$, the attention mass of prototype j on the artifacts of image i, $\mathcal{L}_{\text{art}} = |\mathcal{B}|^{-1} \sum_{i \in \mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}$. Removing it (Ours-noArt) changes accuracy significantly neither on ISIC (ensemble difference 0.007, 95% CI $[-0.004, 0.018]$) nor on any external set ($p \geq 0.058$). Several counterfactual measures are even better without it (e.g., $|\Delta p|$ 0.061 vs. 0.081 for added hair, $p < 10^{-6}$), and the attribution changes little (artifact share $p = 0.36$; pointing $p = 5.7 \times 10^{-5}$; Table V). The explanation gains come from the two core components alone: against PPN-std, PPN and PPN-FA, Ours-noArt has more hot region on the lesion, less on artifacts and fewer flips under all four insertions ($p \leq 5.2 \times 10^{-4}$). Its AUC gain over them (0.010–0.016) is not significant with lesion groups resampled ($p = 0.073$–0.17; image-level DeLong $p = 0.005$ against PPN-std), so the significant gain belongs to the main model. $\mathcal{L}_{\text{art}}$ mainly weakens the response to real bright structures and to dark borders in clinical images (A 0.59 vs. 0.62 and 0.69 vs. 0.75; $p < 0.001$, Table VII); otherwise its effect is small.

I. Robustness to mask quality

Predicted masks could bias the results if a model looked "on the lesion" only by an imperfect mask, so we split the ISIC test set by mask provenance and perturb the lesion masks (Table IX). On the HAM10000 test images, which have expert masks, Ours places the most hot region on the lesion, a *larger* advantage than on the whole test set (+8.8 vs. +2.7 points over the best baseline), so predicted-mask errors do not produce it; its artifact scores are the lowest there too. On the predicted-mask images, Ours has the lowest artifact share, although PPN places slightly more on the lesion (63.5±1.5% vs. 62.4±2.6%).

Eroding or dilating every lesion mask by 5 pixels shifts the lesion fractions of all models alike and keeps the order of the five main models (Ours first); artifact share, pointing and enrichment change by at most 0.011. The provenance guarantee itself depends on the mask border: with masks eroded by 5 pixels, on average 17 of the 20 prototypes of Ours remain on cells with ≥50% lesion (lowest 0.36), all 20 with dilated masks. Ours is also the most accurate of the four main prototype models in every source, with expert (HAM10000) and with predicted training masks (BCN20000, MSK; Table IX).

Too-small predicted masks could let insertions touch the lesion. On exact expert masks, Ours flips the fewest decisions (dark corners 4.4% vs. 7.1% for the CNN and 10.0–10.3% for prototype baselines; rulers 2.3% vs. 3.9% and 4.7–6.1%). On predicted masks it flips the fewest of the prototype models (dark corners 5.7% vs. 10.7–11.5%, CNN 5.5%; rulers 4.4% vs. 7.7–9.7%, CNN 6.7%), with the smallest $|\Delta p|$ in both.

VII. Discussion and Limitations

Lesion attraction changes what the network learns: it becomes more accurate and less dependent on artifacts, with more diverse prototypes. The constrained projection changes what the network shows: it guarantees each explanation's provenance up to mask quality (17 of 20 prototypes with masks eroded by 5 px), at no accuracy cost, for display and audit, but is no mechanism against shortcuts. Our evidence thus supports claims about provenance and robustness to artifacts, not accuracy: the strongest mask-supervised ProtoPNet [12] is as accurate and, across runs, as robust, and the stronger single choice when external generalization matters most. Lesion masks are needed only in training and come from public expert masks of one source, so new dermoscopic data need no annotation; as 7×7 fractions, they tolerate boundary errors (Section VI-I). Anchoring retains the lesion border: 9.3 of 20 Ours prototypes are border cells (lesion fraction 0.5–0.95; baselines and mask-loss variants 1.7–5.0), so ink just outside the border draws some attention (Section VI-D).

Limitations. (i) A gap of about 3 AUC points to the black-box CNN remains; Correia *et al.* [12] found prototype models on par with CNNs, but on ISIC validation data, after a configuration search. (ii) Predicted BCN20000 and MSK masks are validated only indirectly (held-out HAM10000, PH2, subsets), attribution and real-artifact results concern detected artifacts (Section IV), and the lesion share is what $\mathcal{L}_{\text{les}}$ optimizes; insertion, swap and prototype-audit tests need no detector. (iii) Inserted artifacts are synthetic, possibly out of distribution, and kept off the lesion, where anchoring cannot exclude them. Removing real artifacts, a null control and the

test on unedited images mitigate this, but there Ours beats only some models (Section VI-E), possibly partly due to lesion appearance. (iv) The 7×7 grid limits explanation resolution, and class-activation and prototype maps differ in kind, so CNN-prototype attribution comparisons are only indicative; the hot region (top 5%) was not varied, nor were maps randomization-tested [20]. (v) External gains over the prototype baselines are only partly significant. (vi) We studied one backbone and one prototype architecture for a controlled comparison and held out no source in training; both components fit any prototype network with a projection step. Training variability rests on three seeds per configuration (12 paired networks for the projection), whose explanation metrics vary strongly. (vii) Lesion focus is not clinical reasoning: no model favored expert-annotated structures, and no dermatologist rated the prototypes; neither calibration nor skin tone (ISIC is mostly light skin) was assessed; clinical use needs prospective validation in other populations.

VIII. Conclusion

In a same-backbone, leakage-controlled comparison across seven runs, lesion attraction made ProtoPNet somewhat more accurate, also with the background masked, less sensitive to swapped backgrounds, about half as prone to decision flips under inserted dark corners and rulers and less prone under hair and ink; the constrained projection fixed provenance without changing accuracy or robustness in 12 networks. The strongest mask-supervised ProtoPNet was as accurate and, across runs, as robust and generalized better to Derm7pt, but kept all prototypes on lesion-dominated cells in only one of seven runs and learned more redundant prototypes; adding it to our model further weakened the real dark-corner shortcut. **Future work** will test lesion anchoring in other prototype architectures (e.g., PIP-Net [13]) and backbones, combine it with mask supervision while keeping external generalization, use higher-recall artifact detectors and run a reader study.

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